

Scale Fixes Local Errors First: Differential Scaling of Chess Transformer Failures

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Abstract

Scaling reduces how often language models make long-horizon errors, but it is rarely asked which errors it removes. We classify every illegal move made by search-free chess transformers by what made it illegal, using only a rules engine, and measure how the absolute rate of each class falls with model size. Across 18 models spanning 3.4M to 39.9M parameters, errors decidable from the move’s own squares fall with exponent -0.574 , while moves illegal only because they leave the king in check fall with exponent -0.112 : a paired differential of $+0.461$ $[+0.259, +0.599]$. Failures that occur despite correct internal belief about the move’s squares do not fall at all, exponent $+0.023$. The ordering replicates on an independent character-level ladder spanning a 38-fold parameter range, and in OthelloGPT every residual failure is of the non-local kind. A causal account follows: belief is coupled to action locally, a repair edit can correct about two squares, and the class that scale leaves behind is the class local correction cannot reach. The standard intervention metric inverts on the same data. Illegal move rate ranks a norm-matched random edit above correct repair in 18 of 18 models, because hedging lowers it.

Keywords: scaling, failure analysis, world models, state tracking, chess, interpretability, evaluation methodology

1 Introduction

Scaling laws describe how aggregate loss falls with parameters and data (Kaplan et al., 2020; Hoffmann et al., 2022). They say much less about the composition of what remains. A model that makes half as many errors at ten times the size may have removed every kind of error evenly, or removed one kind almost entirely and left another untouched. The two cases recommend different research. If scale removes errors evenly, more scale is the remedy for all of them. If it removes some and not others, the ones it leaves are where effort should go, and interventions aimed at the kind scale already removes are buying something that arrives for free.

Long-horizon reasoning is where this question matters most and is hardest to ask, because in open-ended text there is no ground truth for why a step was wrong. We ask it where there is. A search-free transformer trained on chess move notation produces a move at every position, a rules engine decides whether it is legal, and when it is not, the same engine says why. Some illegal moves are decidable from the two squares the move names: the source square is empty, or holds an opponent’s piece, or the destination holds one of the mover’s own. Others turn on the whole board: the move is geometrically possible for the piece but its path is blocked, or it is pseudo-legal and rejected only because it leaves the king in check. That distinction between local and global illegality is exact, needs no model internals, and is the axis along which we measure scaling.

What we find.

1. **Failure classes scale at different rates, ordered by locality.** Absolute rates of local errors fall steeply with size and global ones slowly. The empty-source class falls 4.01-fold across our ladder and the check class 1.31-fold, a differential exponent of +0.461 that excludes zero in every rung resample. No probe, intervention, or interpretability method enters this result.
2. **It is not an artefact of our models.** The same ordering holds on an independent ladder with different data, notation, and training code, and in a game whose legality is non-local by construction.
3. **The component scale does not remove is the one local correction cannot reach.** Internal belief is causally coupled to action at the squares a move touches, a residual-stream repair can install at most about two squares, and failures that occur despite correct belief at those squares have an exponent indistinguishable from zero.
4. **The usual way of scoring interventions reads this backwards.** Illegal-move rate rewards edits that flatten the output distribution, and on our data it prefers a random direction to the correct repair in every model.

Contributions. A probe-free, rules-grounded measurement of how failure composition changes with scale; its replication across two ladders and two games; a mechanistic account, with a measured capability limit on the causal instrument; and a demonstration that a widely used intervention metric is anti-correlated with the effect it is meant to measure. The probe and edit instruments used in Section 6 are established in prior work (Rathore et al., 2026); here they serve only to explain a result that stands without them.

2 Related Work

Scaling and what it leaves. Aggregate loss follows power laws in parameters, data, and compute (Kaplan et al., 2020; Hoffmann et al., 2022). Decomposing an aggregate into components with separate exponents is the step we take, applied to error classes rather than to loss, in a domain where the classes are defined by an oracle rather than by annotation.

World models in game transformers. Linear probes recover board state from move sequences (Li et al., 2023), with linear structure (Nanda et al., 2023); chess is an established state-tracking testbed (Toshniwal et al., 2022); and board-state probes in a character-level chess model can be intervened on (Karvonen, 2024), later serving as ground truth for dictionary learning (Karvonen et al., 2024). Walker and Lyons (2026) benchmark exact state tracking across architectures at a parameter range close to ours, Li and Jiang (2026) separate state tracking from decision quality, and Pereira and Zuidema (2026) and Harang et al. (2025) document state decay over long generations. State decay and chess state tracking are not our contribution; how the composition of failure changes with size is.

Coupling between probes and behaviour. Balogh and Jelasity (2026) find the fraction of illegal moves legal on a probe-reconstructed board small in large-data models, and a probe gradient nearly orthogonal to the output head. Section 6 reconciles this with a strong causal edit effect: coupling is local, so whole-board measures understate it and square-level ones do not.

Evaluating interventions. Inference-time search and verification (Yao et al., 2023; Lightman et al., 2024; Wang et al., 2023) and causal editing (Meng et al., 2022) are commonly scored by an

end-task error rate. Section 7 shows why that rate can reward perturbations for the wrong reason.

3 Setup

Models. Our ladder has 6 decoder-only transformers from 3.4M to 39.9M parameters, each trained with 3 seeds, for 18 runs. Models read one token per move in UCI notation and are trained on human games, with evaluation on held-out games and no search at inference. Depth and width vary together along the ladder, so a size exponent here is not a pure depth or width exponent (Section 9).

Failure taxonomy. At every position from ply 20 onward we take the model’s top-1 move and ask the rules engine whether it is legal. Each illegal move is assigned to exactly one class, in priority order, by what made it illegal: *empty source* (no piece on the source square), *opponent source* (the source holds the other side’s piece), *own destination* (the destination holds the mover’s own piece), *geometry* (a piece of the right colour is present but cannot reach the destination, or its path is blocked), and *leaves check* (pseudo-legal, rejected only because the king would be in check). The first three are decidable from the two named squares. Geometry depends on intervening squares, and check on the positions of every attacker. Across all 18 runs the residual class that none of these rules catch has a maximum share of 0.0000, which confirms the taxonomy is exhaustive and the parser correct; it is a quality check, not a finding.

Absolute rates, not shares. We report each class as a fraction of all scored positions, its share of illegal moves multiplied by the illegal-move rate. Shares are misleading here: the illegal-move rate itself falls with size, so a class whose absolute rate falls can still grow as a share of what remains. We state this because an earlier draft of this analysis read a rising share as the model getting worse at a class it was in fact getting better at.

Exponents and intervals. A class’s exponent is the slope of log absolute rate on log parameter count, fitted to all runs. Intervals resample the 6 rungs with replacement, since seeds within a rung are not independent draws of the architecture. A differential between two classes is bootstrapped on paired resamples, so its interval is on the difference itself.

Computing environment. Every run in this paper, training and analysis alike, fits on a single NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB of device memory, in a host with 15 GB of physical memory visible to the operating system, under Windows 11 (build 26200), using Python 3.12.10 with PyTorch 2.5.1+cu121 on CUDA 12.1, NumPy 2.4.6, SciPy 1.18.1, and python-chess 1.11.2 as the rules engine. That memory ceiling, not a choice about what was interesting to measure, is what sets the top rung of the ladder at 39.9M parameters, and it is why the external ladders of Section 5 carry the weight they do: they extend the size range past what we can train ourselves. Versions are recorded by `code/record_environment.py` rather than typed here, so they cannot go stale if the work moves to another machine.

4 Differential Scaling

Every class falls with size, so the result is not that larger models get worse at anything. It is that they get better at different things at very different rates (Table 1, Figure 1a). The exponents order

Table 1: Absolute failure rate per scored position at the smallest and largest rung, mean over seeds, and the size exponent with its rung-resampled 95% interval. Classes are ordered by how local the rule deciding them is. The last row is a cross-cutting flag computed with a probe (Section 6), not a sixth disjoint class, and must not be summed with the others.

class	3.4M	39.9M	fold drop	exponent [95% CI]
empty source	0.0582	0.0145	4.01	$-0.574 [-0.748, -0.331]$
own destination	0.0357	0.0112	3.19	$-0.474 [-0.492, -0.452]$
opponent source	0.0093	0.0030	3.05	$-0.437 [-0.586, -0.255]$
geometry	0.0721	0.0443	1.63	$-0.207 [-0.287, -0.150]$
leaves check	0.0776	0.0593	1.31	$-0.112 [-0.153, -0.066]$
all illegal moves	0.2528	0.1323	1.91	$-0.267 [-0.307, -0.216]$
<i>despite correct local belief</i>	0.0388	0.0404	0.96	$+0.023 [-0.005, +0.066]$

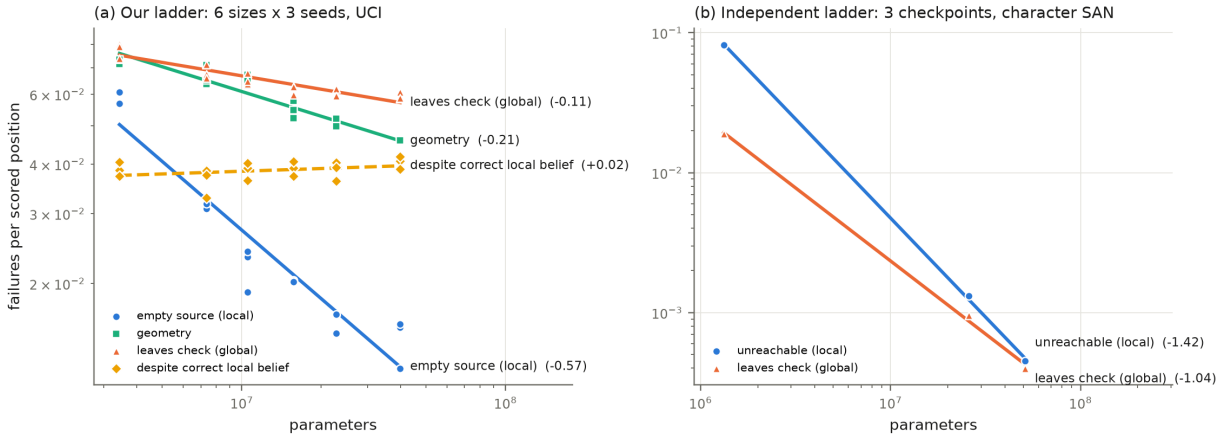


Figure 1: Absolute failure rate against parameter count, log-log, so each line is a power law and its slope, shown in each label, is the exponent. (a) Our ladder; each point is one run. (b) An independent character-level ladder (Section 5); each point is one checkpoint. In both, the local class falls faster than the global one.

themselves by how local the rule is. Empty-source errors fall with exponent -0.574 ; geometry, which depends on the squares between source and destination, with -0.207 ; and check violations, which depend on every attacker’s position, with -0.112 . The paired differential between the check and empty-source exponents is $+0.461 [+0.259, +0.599]$, positive in 100% of rung resamples.

This is a claim about composition, and composition shifts accordingly. The empty-source share of illegal moves falls from 0.230 to 0.110 while the check share rises from 0.307 to 0.448; geometry moves only from 0.285 to 0.335. The shift is substantially one class collapsing into another rather than diffuse rebalancing. Within models, check violations also concentrate at depth: their share rises from early to late game in 6 of 6 rungs, from 0.260 to 0.519 at the largest.

Nothing in this section uses model internals. The measurement is a rules engine and a move counter, which makes it reproducible against any chess model and immune to objections about probe validity.

An illustrative extrapolation. Extending the fitted lines 25-fold beyond our largest model, to one billion parameters, gives check violations at 0.0413 per position against empty-source errors at 0.0023, and failures despite correct local belief at 0.0435. This is far outside the fitted range and we offer it only to show what the exponents imply if they held: the category that scale does not remove would become the largest.

5 External Validity

An independent ladder. Karvonen’s lichess checkpoints (Karvonen, 2024) differ from ours in every respect we could change without training: character-level PGN rather than one token per move, standard algebraic notation rather than UCI, roughly 16M training games rather than ours, and different training code. We use its 1.3M, 25.8M, and 51.0M checkpoints and read the model’s move by greedy character decoding at move boundaries.

Verifying the adapters. Every step between an external model and our measurements fails silently rather than with an error: a wrong vocabulary, a misaligned board, or a missed weight transpose all yield plausible output. Each was therefore checked before use and the checks rerun and recorded. The released weights ship without a tokenizer, so the 32-character vocabulary was reconstructed and accepted only after the model continued 3 of 3 standard openings legally. Board state was aligned to the character before each move and verified three ways at 4,081 sites, by independent replay, by re-parsing the text prefix, and by the boundary character, with 0 disagreements. Checkpoints stored in a different weight layout were converted, and all three trained checkpoints then played 20 of 20 moves legally from the opening, while the randomly initialised control failed at its first move, as it should. A cached decoder used for speed matched the uncached one at all 364 sites tested, the algebraic-notation classifier passed 8 of 8 hand-written cases, and our OthelloGPT forward pass chose a legal move at 672 of 672 positions.

Algebraic notation changes the taxonomy in an instructive way. It never names a source square, so the empty-source class cannot occur: a model that does not name the square cannot place a piece on an empty one. The nearest local class is *unreachable*, no piece of the named type can reach the destination, which also absorbs path blocking. That makes the local class less local than ours and biases the comparison against the result.

The ordering holds (Table 2, Figure 1b). Unreachable errors fall with exponent -1.415 and check violations with -1.044 , a differential of $+0.371$, against $+0.461$ on our ladder. The local share of failures falls from 0.654 to 0.377 while the check share rises from 0.152 to 0.334. Within the smallest checkpoint the check share rises from 0.060 to 0.266 across depth while the unreachable share falls from 0.730 to 0.553. At comparable size this ladder fails far less often than ours, because it was trained on far more data, so only exponents are comparable and never levels.

A game with non-local rules. Othello legality is non-local by construction: a move is legal only if it flanks a line of opponent discs, so it is never decidable from the target square. There is no published Othello ladder, so no exponent can be fitted, and we test only what the claim implies for a single well-trained model: that its residual failures should be the non-local kind. In Li’s synthetic OthelloGPT (Li et al., 2023), across 211,815 positions and 161 failures, 0 are moves onto an occupied square and 160 are moves that flank nothing. The comparison is weaker than it looks, because discs never leave a square: whether a square is occupied reduces to whether it appears in the move history, which requires no state tracking at all. Failures also concentrate late, 6, 32, and 123 across the three move bands.

Table 2: Independent ladder. Absolute rates per scored position. With three checkpoints no uncertainty in the fit itself can be estimated; the intervals propagate only each checkpoint’s sampling noise in class shares and should be read as a lower bound on uncertainty, not as a confidence statement about the slope.

checkpoint	illegal	failures	unreachable	leaves check
1.3M	0.1253	2,836	0.08193	0.01901
25.8M	0.0028	233	0.00132	0.00095
51.0M	0.0012	101	0.00045	0.00040
exponent			$-1.415 [-1.482, -1.365]$	$-1.044 [-1.121, -0.978]$

Table 3: The three mechanism measurements on a model trained on roughly 16M games. Belief consistency is the fraction of illegal moves legal on the probe-decoded board. The AUC on our models is a mean over depth buckets and on Chess-GPT is pooled over depth, which is not the same quantity; only the sign and rough size of the move-touched advantage are compared. Chess-GPT’s move-touched square is the destination, since algebraic notation names no source.

	ours	Chess-GPT, human	Chess-GPT, random play
belief consistency, own board	0.355	0.122	0.188
mismatched board	0.050	0.038	0.061
random move	0.007	0.000	0.003
own / mismatched	7.2	3.2	3.1
AUC, whole board	0.587	0.530	0.485
AUC, move-touched	0.647	0.608	0.552
advantage	+0.060	+0.078	+0.068
edit, state direction	n/a	−0.078	−0.042
edit, norm-matched random		−0.005	−0.005
edit, manipulation check		1.000	1.000

Heavy training. The mechanism in Section 6 rests on three measurements taken on models trained on about a hundred thousand games. We repeated all three on Karvonen’s character-level model trained on roughly 16M games (Table 3). Its board state at depth is better than ours, so the comparison is not against a model whose state was hard to extract: probe accuracy on occupied squares is 0.731 early and 0.315 late, against 0.777 and 0.167 for ours. Human games give only 131 illegal moves, so we also report uniformly random legal play, which lifts the illegal rate from 0.0026 to 0.0231.

All three hold, at reduced size. The model’s illegal moves are legal on its own decoded board far more often than on another position’s, at a ratio of 3.1 against 7.2 for ours. Fidelity at the square the move targets predicts illegality where whole-board fidelity is near chance. And emptying a square in the model’s belief lowers the probability of moves from it, by $-0.037 [-0.050, -0.026]$ beyond a norm-matched random edit, with the decoded square flipped in 1.000 of cases. Heavy training weakens coherent error without removing it.

Table 4: Mechanism summary. Meta rows count models out of 18; repair and fidelity rows are from a single model.

correct beats wrong-target edit, models	18 / 18
correct beats irrelevant-square edit, models	18 / 18
induced single-square corruption, recovered	0.733
random repair	0.253
natural failures, recovered by touched-square repair	0.114
decoded value restored, two-square edit	0.677
decoded value restored, all-square edit	0.304
occupied-square accuracy, before and after all-square edit	0.326 to 0.185

6 Mechanism

Why does scale leave the global class behind? Three measurements, using the probe and edit instruments of Rathore et al. (2026), point to the same answer.

Coupling is local. We repair the model’s belief at the squares its illegal move touches by adding probe-derived occupancy directions to the residual stream, and score the change in a forced-choice preference $r = P(m^*)/(P(m^*) + P(m_{\text{ill}}))$ between the best legal move and the illegal one, which is invariant to uniform flattening of the distribution. Two controls matter. A wrong-target edit installs the same occupancy at an unrelated square; an irrelevant edit spends an identical budget on squares that are wrongly decoded but not touched by the move. Both are matched to correct repair in magnitude and construction, and all contrasts are taken on positions where the two edits changed the output entropy comparably. Across 18 models, correct repair beats wrong-target in 18 (sign test 7.6×10^{-6}) and beats the irrelevant edit in 18 (7.6×10^{-6}), mean advantage +0.039. Only 11 of the irrelevant-edit contrasts exclude zero individually, because matching retains about half the positions per run; the evidence is that none of 18 is negative. Correcting the squares the move depends on changes behaviour more than correcting other wrong squares, which reconciles a weak whole-board coupling (Balogh and Jelasity, 2026) with a strong local one.

Correction reaches about two squares. The repair channel works: when we corrupt a single square ourselves and repair it, the model recovers a legal move in 0.733 [0.699, 0.770] of cases against 0.253 for a random repair. On the model’s own failures, repair at the touched squares restores a legal move in only 0.114, a 6.4-fold gap. Natural failures involve a median of 14 wrongly decoded squares, so we tried repairing all of them. It fails in a specific way (Table 4): a two-square edit restores the decoded value at 0.677 of its targets, while an all-square edit restores 0.304, and full strength per square does no better at 0.279. Whole-board repair lowers occupied-square decoding accuracy from 0.326 to 0.185. The instrument has a capability limit of roughly two squares, whatever the edit magnitude.

What local correction cannot reach. Where decoded belief is already correct at both squares a move touches, a check violation accounts for 0.477 of failures on average against 0.286 for geometry, and is the largest class in 15 of 18 models. Such failures cannot be attributed to the decoded belief being wrong at the move’s squares, and a check violation is decided by squares an edit of two cannot address. Their absolute rate is flat with size, exponent +0.023 [−0.005, +0.066], a differential against empty-source errors of +0.596 [+0.367, +0.761]. An earlier oracle-free attempt

Table 5: Repair conditions on one model (600 positions, 61 games): change in forced-choice preference, fraction of top-1 moves legal after the edit, change in entropy, and how often the target square’s decoded value flipped. The null edit re-installs values already decoded correctly and is zero by construction.

condition	Δr	legal after edit	Δ entropy	decode flipped
correct repair	+0.140	0.147	−0.017	0.826
irrelevant squares	+0.061	0.203	−0.177	0.163
wrong target	+0.053	0.037	−0.007	0.171
random direction	+0.041	0.268	+0.298	0.080
null	0.000	0.000	0.000	0.000

to repair state by re-presenting information the model already attends to also failed, leaving the late-game illegal rate at 0.295 against 0.296 without intervention.

We do not claim these failures are unrepairable in principle. The instrument cannot install a multi-square correction, so whether the residual is genuine policy failure or simply beyond its reach is undetermined, and we recorded the stronger reading as retracted (Appendix B). The claim is narrower and sufficient: the component scale leaves behind is concentrated in exactly the failures that correction at the move’s squares cannot address.

7 The Standard Metric Inverts

Interventions on state are usually scored by whether the model’s output becomes legal. On our repair edits that metric orders the conditions nearly in reverse (Table 5). On one model, correct repair produces the largest preference shift, +0.140, and a random direction the smallest, +0.041; yet after the edit the random direction leaves a legal top-1 move in 0.268 of cases and correct repair in 0.147. The random direction also raises output entropy most, by +0.298. Hedging moves probability off a confidently illegal move and onto whatever else is plausible, some of which is legal.

This is not one model. In 18 of 18 models the random direction produces more legal moves than correct repair while correct repair produces the larger preference shift (sign test 7.6×10^{-6}), and the random direction has the best legal-move rate of all four edits in 16 of them. Its advantage in legal moves over correct repair ranges from +0.048 to +0.143.

The same artefact affects our own preference metric if uncontrolled. The raw advantage of correct repair over wrong-target is +0.087; restricted to positions where both edits changed entropy comparably it is +0.036 [+0.011, +0.058], 2.4 times smaller. The edit also responds gradedly to strength: preference rises from +0.042 at a quarter of calibrated strength to +0.140 at calibrated strength, then turns to −0.046 and −0.091 at double and quadruple, a rise-then-collapse that is harder to explain as generic damage than a monotone decline would be. Studies that score state interventions by illegal-move rate, or by any error rate a flatter distribution can lower, may be measuring hedging.

8 Discussion

The practical content is about where effort goes. The class that does not fall with size is the one defined by constraints spanning the whole board, and on our evidence it is also the class that correcting the model’s belief about a move’s own squares does not reach. We read this, as an interpretation we have not tested directly, as a caution for interventions such as auxiliary board

losses and state supervision: they aim at the model’s knowledge of where pieces stand, and the failures most directly attributable to that knowledge are the ones scale is already removing. An intervention that lowers total error while leaving the global class flat may have improved only what scale improves anyway, and should be evaluated by its effect on that class.

This also bounds what an intervention study can show. A training comparison that targets state-attributable failure is adequately powered on our data, detecting a change of 19% of that component with seed variation contributing little (relative spread 0.028), but the component is a small share of what goes wrong, so a detectable improvement is a small behavioural one.

The local and global distinction is not specific to chess. Wherever a checker can say not only that a step is wrong but which constraint it violated, program execution, spreadsheet recalculation, and formal proof among them, the same decomposition is available, and the question of which violations scale removes can be asked without any access to model internals.

9 Limitations

Depth and width are confounded. Both ladders grow depth and width together, so an exponent in parameters does not isolate either.

Three external checkpoints. The independent ladder supports directional agreement, not a second estimate of the slope; its intervals omit fit uncertainty.

Notation changes the classes. Algebraic notation removes the empty-source class, and its nearest local analogue absorbs path blocking.

Othello cannot test scaling. No ladder exists, and its local class requires no state tracking, so it is a consistency check rather than a replication.

The residual is unattributed. The repair instrument corrects about two squares, so whether failures despite correct local belief reflect policy or unreachable state is undetermined.

Probe-based rows depend on a linear probe. A nonlinear decoder of four times the capacity gains only +0.011 to +0.033 across depth (Appendix A), so decoder weakness does not explain decay, but the probe-based rows inherit its approximation. The headline result uses no probe.

One game family. Chess and Othello both carry exact oracles, which is what makes them measurable and what makes them unrepresentative of open-ended reasoning.

10 Conclusion

Scale does not remove long-horizon errors evenly. In chess transformers it removes errors decidable from a move’s own squares several times faster than errors decided by the whole board, the ordering replicates on an independent model family, and the failures that occur despite correctly decoded belief at the move’s squares do not decline at all. The measurement needs only a rules engine. What it suggests is that the failures worth studying at scale are the ones defined by global constraints, and that interventions should be judged by their effect on those, measured with a metric a flatter output distribution cannot improve.

Declarations

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Data availability statement. Trained checkpoints, analysis code, and every result file cited by the manuscript are available from the corresponding author on reasonable request. External checkpoints are public.

AI usage declaration. During preparation the authors used a generative AI tool to assist with analysis code and language editing. The authors reviewed and edited all output and take full responsibility for the content.

A Probe capacity

Every fidelity curve in this line of work is read with a linear probe, which could confound information leaving the representation with information becoming nonlinearly encoded. On identical activations and budgets, a two-layer probe whose hidden layer is four times the residual width improves occupied-square accuracy by between +0.011 and +0.033 across depth, with no tendency to widen. The same nonlinear probe reaches 0.000 on a control task with shuffled boards and 0.015 on a randomly initialised model in the deepest bucket, so there is no memorisation headroom for extra capacity to exploit. Decay in these curves is therefore not explained by the weakness of a linear decoder, though a single alternative decoder cannot rule out every nonlinear encoding.

B Integrity record

The analysis was pre-registered (SHA-256 prefix 28b68e5200dc0dba). Three amendments changed reported claims and are stated here in full. First, the frozen entropy-matching procedure was vacuous, since weighting strata by their own counts reproduces the overall mean; it was replaced by matching on positions after point estimates were seen, and the replacement reduced the headline effect. Second, a prior that about a third of illegal moves would be restored to legality by repair was mis-specified, by conflating whether an error is explicable by wrong belief with whether repairing belief is sufficient to change behaviour; the observed rate was far lower. Third, a conclusion that the failures repair could not fix are genuine was retracted once whole-board repair was shown not to install a board, which is why Section 6 leaves that residual unattributed.

Two further corrections changed how a result was stated rather than a reported value. The scaling result was first framed in terms of shares of failures, and that framing was stated before it was found to imply that models grew worse at a class they were in fact improving at; it was revised to absolute rates, which is the form reported here. The metric comparison was first reported on one model and is now reported across all of them. Analysis corrections that changed no reported claim, including a move-touched definition that scored illegal algebraic moves at arbitrary squares and edit intervals clustered on too few games, are recorded in the analysis repository’s version history.

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